Exploring Security and Advanced Concepts Activity Book

Data Modeling for Qlik Sense

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Exploring Security and Advanced Concepts Activity Book

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Activity 1: ODAG in action

Summary:

- Explore an On-demand selection app.
- Generate a template app data slice from a selection in the main app.

Qlik Sense apps: Demo - ABC Selection app, Demo - ABC Template app

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Challenge

Use pre-created **ODAG** demo apps to perform analysis on a sliced version of the entire dataset.

Hints

- Make selections in the **Selection app** and generate the **Template app**.
- Open the **Template app** to analyze the sliced data.

Step-by-step guide

1. From the course collateral, upload the two files for the activity to the Hub:
   - *Demo - ABC ODAG Selection app.qvf*
   - *Demo - ABC ODAG Template app.qvf*

   Qlik Sense Client-managed: The two apps required for the activity are located in the **DA - Exploring security and advanced concepts** stream . They are already linked and ready to be tested.

2. Review the two apps. The **Demo - ABC Selection app** contains high-level aggregations of the entire data set. The **Demo - ABC Template app** is used as a template for the details app. It only contains visualizations and script for more granular analysis of the **Selection app**. It does not contain any data.

3. In the **Demo - ABC Selection app**, edit the **Order Details** sheet.
4. From the **Advanced options** view, if available, open **App navigation links** from the **Assets** panel. If there is already an *Order Details* app navigation link in the panel, use the menu button and delete the link. In addition, delete the navigation point from the bottom of the sheet.

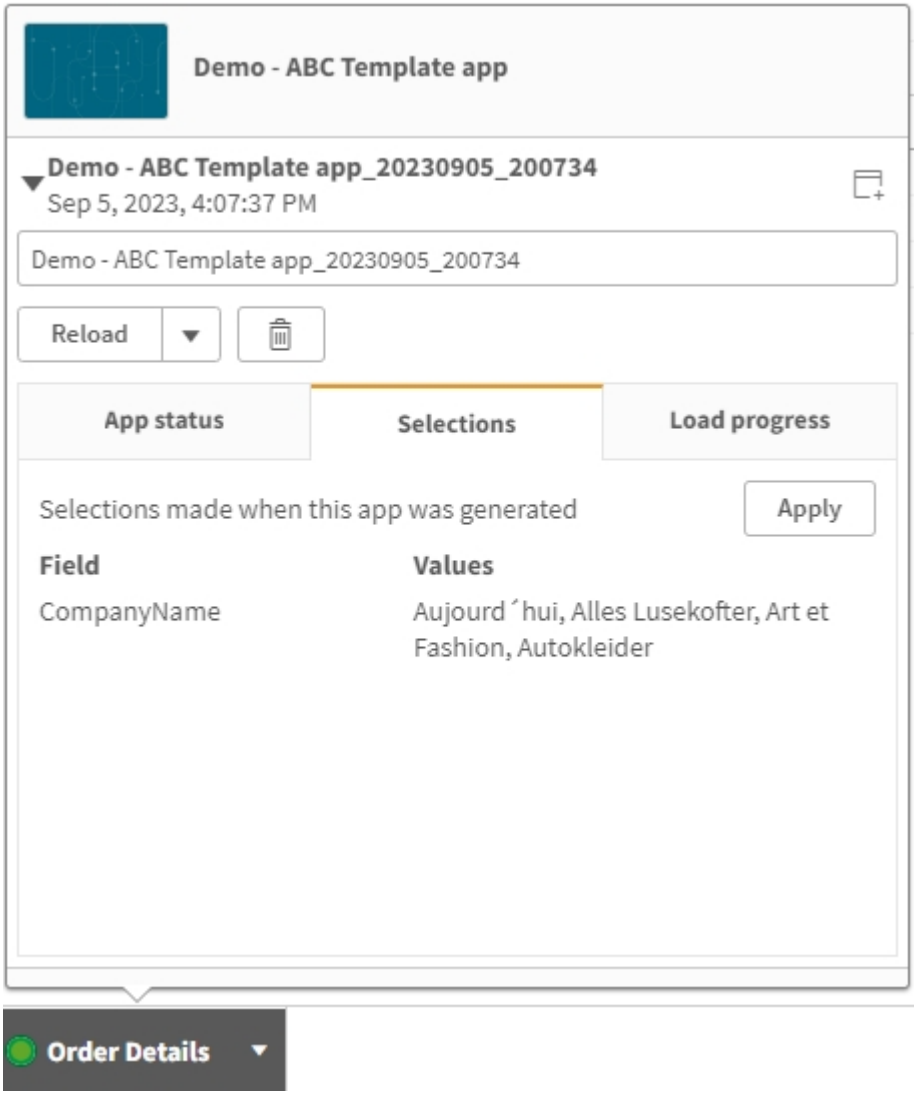
   Note: The **App navigation links** rely upon unique app identifiers. These identifiers are re-generated when you import an app. Therefore, importing the .qvf files will effectively break the app nav links between the two apps.

5. Use the **Create new** button to set up a new navigation link.
6. Populate the settings in the dialog box as follows:
   - **Name**: *Order Details*
   - **Template app**: *Demo - ABC Template app*
   - **Expression**: *Count(Distinct OrderID)*
   - **Maximum row count**: *150*
   - **Generated apps**: **Maximum number**: *5*
   - **Retention time**: *Never expires*
   - **Default view when opened**: *App overview*
7. Select **Create** to confirm the new navigation link settings, then drag the new *Order Details* nav point to the **App navigation** area for the sheet.
8. Exit sheet edit mode, and return to the Hub.
9. Re-open the **Demo - ABC Selection app**. The first sheet presents an overview on how **ODAG** works. Review the content. Use the *Click here to start* link at the bottom of the sheet or use the arrows to navigate to the next sheet.
   The **Order Details** sheet in the app contains high-level aggregations of the entire dataset. To drill-down to more granular information using **ODAG**, make selections to reduce the *# Orders* to less than *150*. Then, generate the new details app. Use the instructions on the sheet to assist in this process. In this **Demo** app, 150 is the value we manually defined to be the maximum limit. This number can be adjusted accordingly.

   Note: Notice that while you are filtering your data, if the *# Orders* is under 150, the donut-shaped icon at the bottom left corner of the sheet becomes a green dot. If it becomes greater than *150*, it will remain a donut with only a partial green fill.

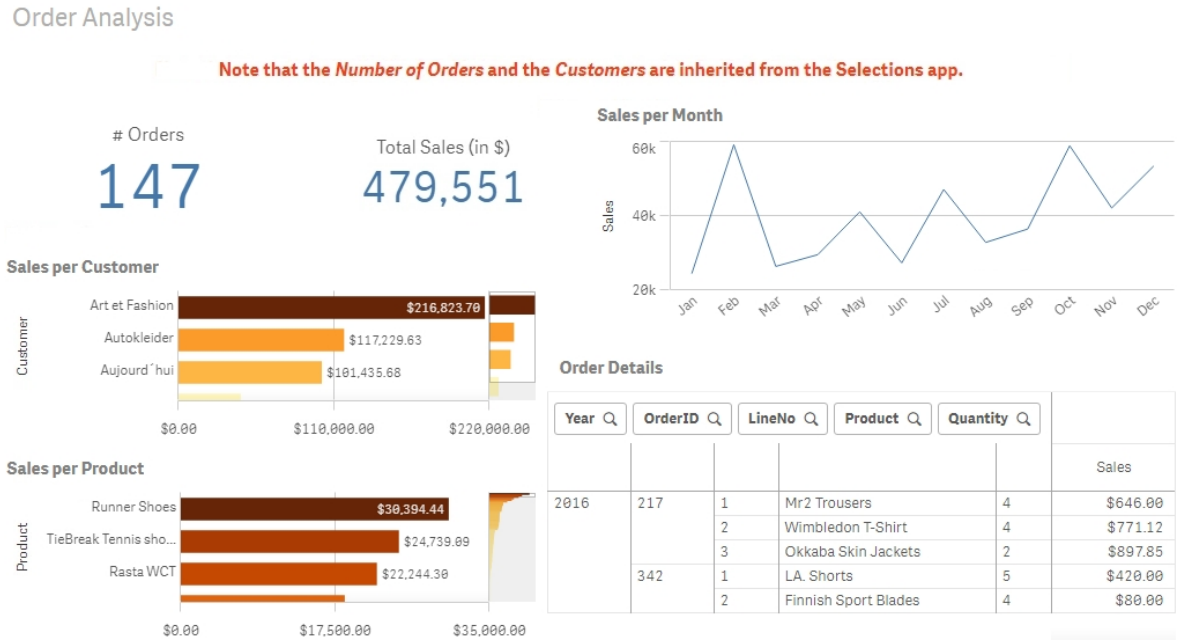
10. Select only the first five values in the *CompanyName* filter pane. Notice that the total *# Orders* for the first five values is more than *150*, meaning that the details app cannot be generated.
11. From the same filter pane, select only the first four values. The *# Orders* value is now less than *150* and the circular icon has turned into a green dot. This indicates that the details app can be generated.
12. Click on the **Order Details** green icon and select **Generate new app**. The current app uses script to send the selected values as parameters and reloads the **Demo - ABC Template app**. The parameters will be used to filter the amount of data loaded by the details app. To review the reload progress, click on the black arrow next to the details app (**Demo - ABC Template**) to expand the section.
13. Review all the available options and navigate to the **Selections** and **Load progress** tabs. Verify the values that passed as parameters, from the *CompanyName* field to the details app, are the ones you selected.

(Click on the image below to enlarge)



14. To open the details app, click on the icon next to the app name.
15. The details app opens and you can see the **Order Analysis** sheet. Review the visualizations and verify that the sheet performs analysis only for the four values you selected.

Solution



Activity 2a: Apply basic Section Access (SaaS)

Summary:

- Implement a Section Access security to an app.
- Validate basic security is working by logging in as different users.

Qlik Sense app: ABC Sales Data / ABC Sales Data_SA1_<user>

Challenge

In the **ABC Sales Data** app, add **SECTION ACCESS** to provide extra application level security. To achieve that, use a number of pre-existing users as per the table below:

USERID	Name
Qlik.Bing.Martin@protonmail.com	Bing Martin
Qlik.Frank.Roll@protonmail.com	Frank Roll
Qlik.Ingrid.Hendrix@protonmail.com	Ingrid Hendrix
Qlik.Joan.Callins@protonmail.com	Joan Callins

Note: The password for all users is *Training1*.

To add **SECTION ACCESS** to your app, complete the following tasks:

- Open and update the file *SectionAccess.txt* from the student collateral, adding your own email address.
- Within the **Personal** space, duplicate the **ABC Sales Data** app and rename the copy *ABC Sales Data_SA1* and suffix it with _<your name>.
- In the beginning of your script, add a new script section named *Security.* Add a script to load the above file into the **SECTION ACCESS** area.
- Load your data.
- Move the app to the **DMQS** space.
- Login as *FRL* and then as *IHX.* Observe the different access levels.

Hints

- Ensure the *SectionAccess.txt ADMIN* user correctly reflects your own Qlik ID email.

Step-by-step guide

1. From the student collateral, use a text editor to open the *SectionAccess.txt* file.
2. Replace the *your.email@address*, with your own Qlik ID login email:

```
ACCESS, USER.EMAIL
ADMIN, your.email@address
USER, qlik.bing.martin@protonmail.com
USER, qlik.frank.roll@protonmail.com
```

3. Save and close the file.
4. Within the **Personal** space, open the actions menu for the **ABC Sales Data** app and use the **Duplicate** option. After a moment, a new app named **Copy - ABC Sales Data** is created.

Note: It is recommended to **Duplicate** the app before applying **Section Access** to avoid making logical errors while developing the security.

5. Rename this new copy to *ABC Sales Data_SA1_* and your name as a suffix.
6. Open the app and navigate to the **Data load editor**. Create a new script section named *Security.* Move the new section to the top of the section list.
7. In the **Security** section, add the following statement:

```
SECTION ACCESS;
```

8. Add a few carriage returns, then use the **Select data** option for the **DataFiles** connection. Drag the *SectionAccess.txt* file onto the dialog to upload it.
9. Select the file from the dialog box to open it.
10. Make sure that the *Embedded field names* option is selected for the table and insert the script. This loads information for the access level and user ID.
11. After this script, add the following line:

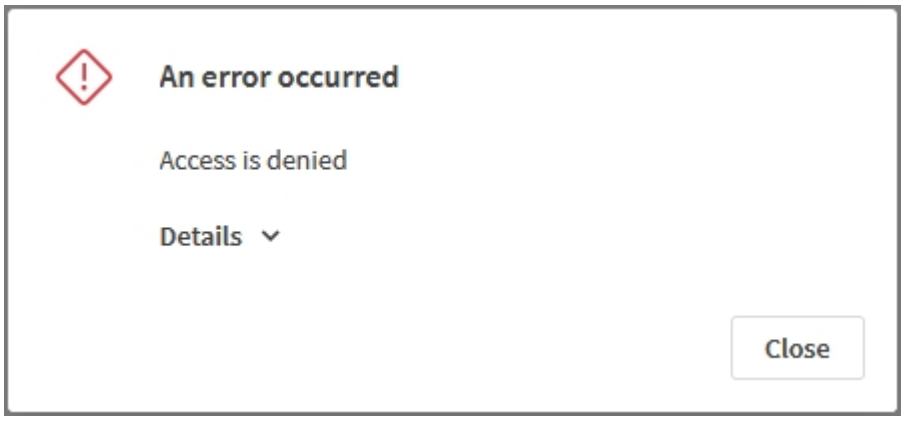
```
SECTION APPLICATION;
```

12. Load the data to add the security information, then close the **Data load progress** dialog box.
13. Return to the **Personal** space in the hub.
14. For the new **ABC Sales Data_SA1** app, use the actions option to open the menu, and select **Move**.
15. Select the Space: **DMQS** and proceed with the **Move**, navigating to the space, as required.

> **Note:** The remaining steps of this activity involve logging in-and-out of **Qlik Sense Enterprise SaaS**, from a single browser. If a second browser is available, feel free to leverage it, i.e. remain logged in as your main account in one browser, but utilize an additional browser to log in as the test users. If you do so, ensure you use the URL provided by the instructor to access the specific cloud tenant you are using for training.

16. Click the **User profile menu** to open it, and select **Log out**.
17. Sign back into the cloud tenant as *Frank Roll*, using *Qlik.Frank.Roll@protonmail.com* as the user ID and *Training1* as the password.
18. Ensure the **FR** credentials appear as the profile on the **Hub** navigation bar.
19. Navigate to the **DMQS** space, via the **Catalog**.
20. Click the **ABC Sales Data_SA1** app. Can you open it?
21. Close the app and log Frank Roll out of the tenant.
22. Repeat the previous steps to log on as a different user: *Qlik.Ingrid.Hendrix@protonmail.com*. This user, *Ingrid Hendrix*, receives an **Access is denied** message when trying to open the app. She has no user access in the security file.
23. Use the browser **Back** button to return to the Hub.
24. Log Ingrid out of the tenant.

Solution



Activity 2b: Apply basic Section Access (QSoW)

Summary:

- Implement a Section Access security to an app.
- Validate basic security is working by logging into the Hub as different users.

Qlik Sense app: ABC Sales Data / ABC Sales Data_SA1

Challenge

In the **ABC Sales Data** app, add **SECTION ACCESS** to provide extra application level security. To achieve that, use a number of local users as per the table below:

USERID	Local user name
ABC\STUDENT	Student—the account you have been using
ABC\BMN	Bing Martin
ABC\FRL	Frank Roll
ABC\IHX	Ingrid Hendrix
ABC\JCS	Joan Callins

Note: The password for all users is *Training1*.

To add **SECTION ACCESS** to your app, complete the following tasks:

- Open and examine the file *SectionAccess.txt*.
- In the **Qlik Sense Hub**, duplicate the **ABC Sales Data** app and rename the copy *ABC Sales Data_SA1*.
- In the beginning of your script, add a new script section named *Security.* Add a script to load the above file into the **SECTION ACCESS** area.
- Load your data. Using the **Qlik Management Console (QMC)**, publish the app to the **Everyone** stream.
- Login as *FRL* and then as *IHX.* Observe the different access levels.
- Close all **Google Chrome** browser windows.

Hints

- The file that contains the security information is stored in the following location: **C:\Qlik Sense Data\TextFiles**.

Step-by-step guide

1. Examine the data in the *SectionAccess.txt* file.
2. Open the **Qlik Sense Hub** and right-click the **ABC Sales Data** app for the menu option to **Duplicate**. Using this option, a new app named **ABC Sales Data (1)** is created.

Note: It is recommended to **Duplicate** the app before applying **Section Access** to avoid making logical errors while developing the security.

3. Refresh the app list and edit this app to rename it to *ABC Sales Data_SA1*.
4. Open the **Data load editor** from the top navigation. Create a new script section named *Security.* Move the new section to the top of the section list.
5. In the **Security** section, add the following statement:

```
SECTION ACCESS;
```

6. After the previously inserted statement, place the cursor in the script and use the **TextFiles** connection to load the *SectionAccess.txt* table.
7. Make sure that the *Embedded field names* option is selected for the table and insert the script. This loads information for the access level and user ID.
8. After this script, add the following line:

```
SECTION APPLICATION;
```

9. Load the data to add the security information and then open the **QMC**.
10. On the **Apps** page, select and publish the **ABC Sales Data_SA1** app to the **Everyone** stream.
11. From the Windows **Start** button, right-click on the **Google Chrome** icon, and select **Run as different user**.
12. Log in as *Frank Roll*, using *FRL* as the user ID and *Training1* as the password.
13. When **Google Chrome** opens, make sure you are viewing **https://qse/hub/**.
14. In the **Qlik Sense Hub**, notice how the **FRL** credentials appear as the profile on the top bar, along the left side. Select the **Everyone** stream.
15. Click the **ABC Sales Data_SA1** app. Can you open it?
16. Close the browser window and repeat the previous steps to log on as a different user: *IHX*. This user, *Ingrid Hendrix*, receives an **Access is denied** message. She has no user access in the security file.
17. Close all the new **Google Chrome** browser windows.

Solution

Activity 3a: Data reduction with Section Access (SaaS)

Summary:

- Create an app-level security layer via Section Access security.
- Investigate the different levels of row and column-level access for users.

Qlik Sense app: ABC Sales Data / ABC Sales Data_SA2_<user>

Challenge

Add **Section Access** with data reduction using the same users as before. To achieve this, complete the following tasks:

- Examine the *DataReductionSA.txt* and *DataReductionLink.txt* files. Update *DataReductionSA.txt* to use your QlikID tenant email.
- In the **Hub**, open the original **ABC Sales Data** app and add a *DivisionName* **Filter** control in the **Employees** app sheet, then make the sheet public.
- In the **Hub**, duplicate the app and rename the new app *ABC Sales Data_SA2* and suffix it with _<yourname>
- At the top of the script sections, add a new section, named *Security*. Load the *DataReductionSA.txt* above the **SECTION ACCESS** area.
- Add the *DataReductionLink.txt* after the **SECTION APPLICATION** statement, on the same section.
- Load your data. Move the app into the **DMQS** space.
- Log on as *qlik.frank.roll@protonmail.com,* then *qlik.ingrid.hendrix@protonmail.com,* and then to *qlik.bing.martin@protonmail.com.* Observe the different access levels. See how the data is reduced for the different users.

Hints

- Ensure the *DataReductionSA.txt ADMIN* user correctly reflects.your own Qlik ID email.
- The first security file contains the **SECTION ACCESS** table. This table includes a *SA_LINK* field, used for the reduction of the available rows. It also includes an *OMIT* column to remove the *JobTitle* field. This first file is *DataReductionSA.txt*.
- The second file is loaded as part of **SECTION APPLICATION**. It acts as a link table between the **SECTION ACCESS** table and the data model, reducing the available rows based on *DivisionName*. This second file is *DataReductionLink.txt*.

Step-by-step guide

1. From the student collateral, use a text editor to open the *DataReductionSA.txt* file.
2. Replace the *USER.EMAIL* field value for the *ADMIN* user (your.email@address), with your own Qlik ID login email:

```
ACCESS, USER.EMAIL, SA_LINK, OMIT
ADMIN, your.email@address, *,
USER, qlik.joan.callins@protonmail.com, EMEA,
USER, qlik.frank.roll@protonmail.com, NA,
USER, qlik.ingrid.hendrix@protonmail.com, SA, JobTitle
```

```
USER, qlik.ingrid.hendrix@protonmail.com, NA, JobTitle
USER, qlik.bing.martin@protonmail.com, *,
```

3. Save the file.
4. Examine both the *DataReductionSA.txt* and *DataReductionLink.txt* files, then close them.
5. In Qlik Sense, navigate to the **Personal** space and make sure you are logged on as your main Qlik ID user.
6. In the **ABC Sales Data** app, open the **Employees** sheet and enable edit mode.
7. Make space for a new **Filter** control and add the *DivisionName* field onto the sheet as the new **Filter**.
8. Disable the edit mode and make the **Employees** sheet public.
9. Navigate back to the **Personal** space in the **Hub**.
10. Duplicate the **ABC Sales Data** app and rename this new copy to *ABC Sales Data_SA2_* and your name as a suffix.
11. Open *ABC Sales Data_SA2* and navigate to the **Data load editor**. Create a new script section named *Security*. Move the new section to the top of the section list.
12. In the **Security** section, add the following statement:

```
SECTION ACCESS;
```

13. Add a few carriage returns, then use the **Select data** option for the **DataFiles** connection. Drag the *DataReductionSA.txt* file onto the dialog to upload it.
14. Select the file from the dialog box to open it.
15. Make sure the *Embedded field names* option is selected and insert the script. This loads information for the access level, user ID, link to second table, and the *OMIT* fields column. Insert the script.
16. After this script, add the following line:

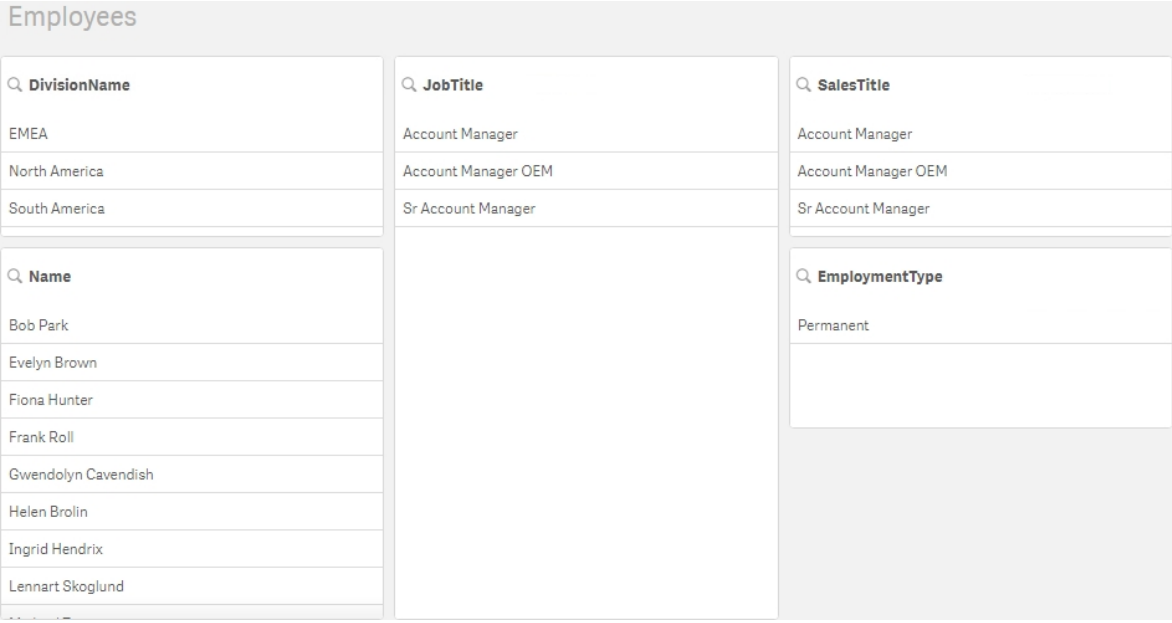
```
SECTION APPLICATION;
```

17. Add a few more carriage returns, then use the **Select data** option for the **DataFiles** connection again. Drag the *DataReductionLink.txt* file onto the dialog to upload it.
18. Select the file from the dialog box to use it.
19. Make sure that the *Embedded field names* option is selected and insert the script. This loads the link table information associating each user to the relevant division they have access to.
20. Load the data to add the security information and close the app to return to the **Personal** space in the **Hub**.
21. For the new **ABC Sales Data_SA2** app, use the **More actions** option to open the menu, and select **Move**.
22. Select the Space: **DMQS** and proceed with the **Move**, navigating to the space, as required.

Note: The remaining steps of this activity involve logging in-and-out of **Qlik Sense Enterprise SaaS**, from a single browser. If a second browser is available, feel free to leverage it, i.e. remain logged in as your main account in one browser, but utilize an additional browser to log in as the test users.

23. Click the **User profile menu** to open it, and select **Log out**.
24. Sign back into the cloud tenant as *Frank Roll*, using *Qlik.Frank.Roll@protonmail.com* as the user ID and *Training1* as the password.
25. Ensure the **FR** credentials appear as the profile on the **Hub** navigation bar.
26. Navigate to the **DMQS** space, via the **Explore** option.
27. Open the **ABC Sales Data_SA2** app and select the **Employees** sheet. *Frank* has access to all fields, but only data associated to *North America*.
28. Return to the **Hub** and log Frank out.
29. Log back on as *Qlik.Ingrid.Hendrix@protonmail.com* with the password *Training1*. Open the **ABC Sales Data_SA2** app again. Select the **Employees** sheet. *Ingrid* only has access to *North America* and *South America*. She cannot see *JobTitle* data because this field has been omitted.
30. Note that the *SalesTitle* is still available. Consider how you can also omit this field for *Ingrid*.
31. What can you see when you log in as *Bing Martin*, using *Qlik.Bing.Martin@protonmail.com* as the user ID? Why? (see *Solution image*)
32. Use the browser **Back** button to return to the Hub, and log Bing out.

Solution



Activity 3b: Data reduction with Section Access (QSoW)

Summary:

- Create an app-level security layer via Section Access security.
- Investigate the different levels of row and column-level access for users.

Qlik Sense app: ABC Sales Data / ABC Sales Data_SA2

Challenge

Add **SECTION ACCESS** with data reduction using the same users as before. To achieve this, complete the following tasks:

- Examine the *DataReductionSA.txt* and *DataReductionLink.txt* files.
- In the **Qlik Sense Hub**, open the **ABC Sales Data** app and add a *DivisionName* **Filter** control in the **Employees** app sheet.
- In the **Qlik Sense Hub**, duplicate the app and rename the new app *ABC Sales Data_SA2*.
- At the top of the script sections, add a new section named *Security*. Add the *DataReductionSA.txt* above the **SECTION ACCESS** area.
- Add the *DataReductionLink.txt* after the **SECTION APPLICATION** statement, on the same section.
- Load your data. Publish the app using the **Qlik Management Console (QMC)** into the **Everyone** stream.
- Log on as *FRL,* then *IHX,* and then to *BMN.* Observe the different access levels. See how the data is reduced for the different users.
- Close all **Google Chrome** browser windows.

Hints

- The files that contain the security information are stored in the following location: **C:\Qlik Sense Data\TextFiles**.
- The first security file contains the **SECTION ACCESS** table. This table includes a *SA_LINK* field, used for the reduction of the available rows. It also includes an `OMIT` column to remove the *JobTitle* field. This first file is *DataReductionSA.txt*.
- The second file is loaded as part of **SECTION APPLICATION**. It acts as a link table between the **SECTION ACCESS** table and the data model, reducing the available rows based on *DivisionName*. This second file is *DataReductionLink.txt*.

Step-by-step guide

1. Open and examine the *DataReductionSA.txt* and *DataReductionLink.txt* files.
2. Open the **Qlik Sense Hub** and make sure you are logged on as *Student.*
3. In the **ABC Sales Data** app, open the **Employees** sheet and enable the edit mode.
4. Make space for a new **Filter** control and add the *DivisionName* field onto the sheet as the new **Filter**.
5. Disable the edit mode and navigate back to the **Hub**.
6. Duplicate the **ABC Sales Data** app and rename it to *ABC Sales Data_SA2*.
7. Navigate to the **Data load editor**. Create a new script section named *Security.* Move the new section to the top of the section list.
8. In the **Security** section, add the following statement:

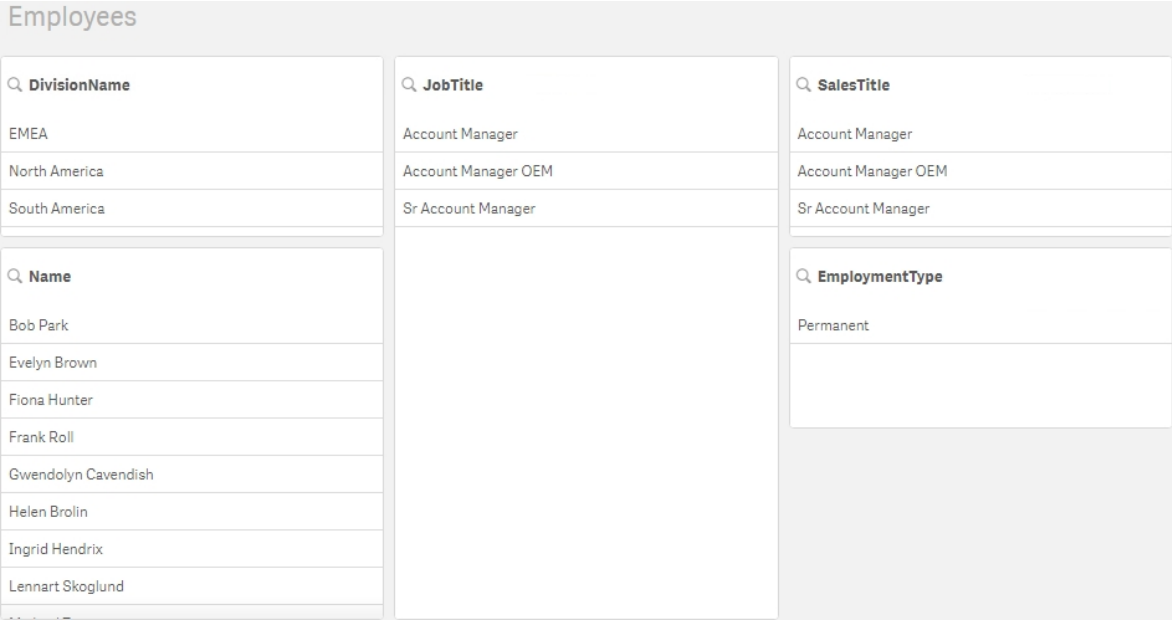
```
SECTION ACCESS;
```

9. After inserting the **Section Access** statement, use the **TextFiles** data connection to load the *DataReductionSA.txt* file.
10. Make sure the *Embedded field names* option is selected and insert the script. This loads information for the access level, user ID, link to second table, and the `OMIT` fields column.
11. After this script, add the following line:

```
SECTION APPLICATION;
```

12. After inserting the **Section Application** statement, use the **TextFiles** data connection again, to load the *DataReductionLink.txt* file.
13. Make sure that the *Embedded field names* option is selected and insert the script. This loads the link table information associating each user to the relevant division they have access to.
14. Load the data to add the security information and open the **QMC**.
15. In the **Apps** page, select and publish the **ABC Sales Data_SA2** app into the **Everyone** stream.
16. From the Windows **Start** button, right-click on the **Google Chrome** icon, and select **Run as a different user**.
17. Log in as *Frank Roll*, using *FRL* as the user ID and *Training1* as the password.
18. When **Google Chrome** opens, make sure you are viewing **https://qse/hub/**.
19. When the **Qlik Sense Hub** opens, confirm the **FRL** credentials appear on the top bar. Select the **Everyone** stream.
20. Open the **ABC Sales Data_SA2** app and select the **Employees** sheet. *Frank* has access to all fields, but only data associated to *North America*.
21. Log on as *Ingrid Hendrix* this time, with the user name *IHX* and the password *Training1*. Open the **ABC Sales Data_SA2** app again. Select the **Employees** sheet. *Ingrid* only has access to *North America* and *South America*. She cannot see *JobTitle* data because this field has been omitted.
22. Note that the *SalesTitle* is still available. Consider how you can also omit this field for *Ingrid*.
23. What can you see when you log in as *Bing Martin*, using *BMN* as user ID? Why? (see *Solution image*)
24. Close all the **Google Chrome** browser windows.

Solution



Activity 4a: Omit a group of fields (SaaS)

Summary:

- Modify app-level security to omit groups of fields.
- Analyze sheet filter objects to verify the results.

Qlik Sense app: ABC Sales Data_SA2 / ABC Sales Data_SA3_<user>

Challenge

In the **ABC Sales Data_SA3** app, omit a group of fields by doing the following:

- Create a copy of the *DataReductionSA.txt* and rename it to *DataReductionSA_OPT.txt*.
- In the new file, replace *OMIT* with *OMIT_GROUP* and all the *JobTitle* instances with *OMIT_GROUP1*.
- In the **Hub**, open the **DMQS** space and duplicate the **ABC Sales Data_SA2** app.
- Change the name to *ABC Sales Data_SA3*, suffixed with _<your name>.
- In the **Security** script section, modify the **SECTION ACCESS LOAD** statement to load from the new file.
- Add a second **LOAD** statement containing table columns *OMIT_GROUP* and *OMIT*, and fields *JobTitle*, *SalesTitle*, and *EmploymentType*, associated in the same group, *OMIT_GROUP1*.
- Load your data.
- Log on as *IHX*. Observe the fields the user has access to in the **Employees** sheet.

Hints

- Ensure the *DataReductionSA_OPT.txt* **ADMIN** user correctly reflects.your own Qlik ID email.

Step-by-step guide

1. Examine the *DataReductionSA.txt* you loaded in the previous activity.
2. Duplicate this file and rename the copy as *DataReductionSA_OPT.txt*. Open the file to edit it.
3. In the file, replace *OMIT* with *OMIT_GROUP*. Also, replace all the *JobTitle* instances with *OMIT_GROUP1*. Then, **Save** and close the file.
4. Return to the **Hub** and make sure you are logged on as your primary Qlik ID. Go to the **DMQS** space.
5. Duplicate the **ABC Sales Data_SA2** app which contains your name. Rename the copy that appears in the space to *ABC Sales Data_SA3_*<yourname>.
6. Open the new app, navigate to the **Data load editor** and locate the **Security** section.
7. In the **Data connections** panel, expand the **Personal** space, if required, and use the **Select data** option for the **DataFiles** connection. Drag the *DataReductionSA_OPT.txt* file onto the dialog to upload it.
8. Click **Cancel** to close the dialog box.
9. Within the script, replace the name of the file *DataReductionSA.txt* with *DataReductionSA_OPT.txt* in the first **FROM** clause.
10. In the **SECTION ACCESS LOAD**, replace the *OMIT* field with *OMIT_GROUP*.
11. Between the **SECTION ACCESS** table and the **SECTION APPLICATION** statement, add the following script:

```
LOAD * INLINE [
    OMIT_GROUP, OMIT
    OMIT_GROUP1, JobTitle
    OMIT_GROUP1, SalesTitle
    OMIT_GROUP1, EmploymentType
];
```

Copy to clipboard

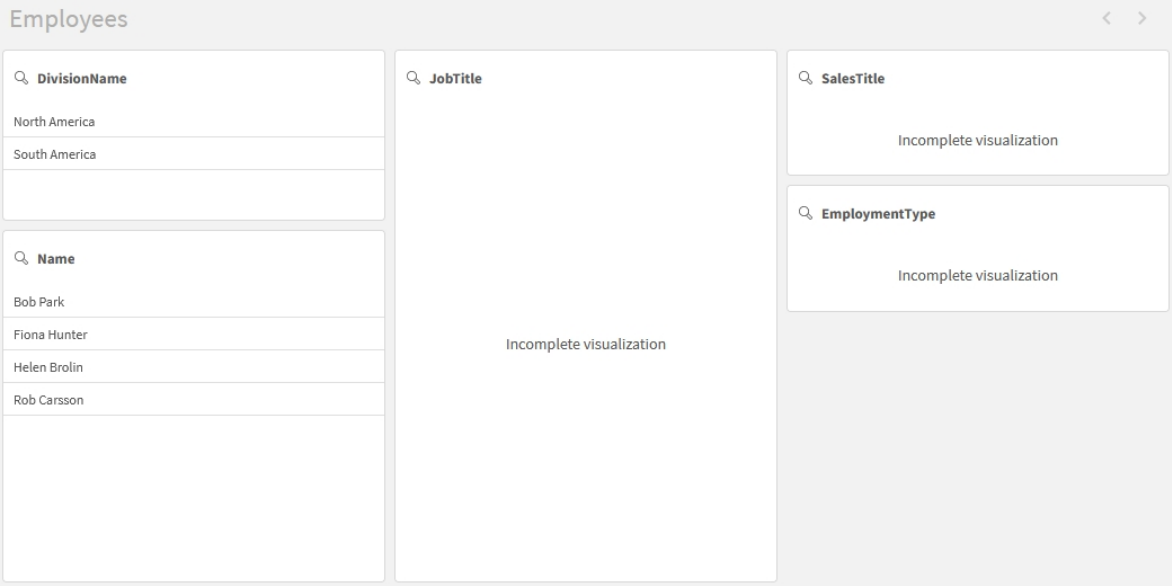
12. Load the data to add the security information to the app and return to the **Hub**, displaying the **DMQS** space

> **Note:** The remaining steps of this activity involve logging in-and-out of **Qlik Sense Enterprise SaaS**, from a single browser. If a second browser is available, feel free to leverage it, i.e. remain logged in as your main account in one browser, but utilize an additional browser to log in as the test users.

13. Click the **User profile menu** to open it, and select **Log out**.
14. Sign back into the cloud tenant as *Ingrid Hendrix*, using *Qlik.Ingrid.Hendrix@protonmail.com* as the user ID and *Training1* as the password.
15. Ensure the **IH** credentials appear as the profile on the **Hub** navigation bar.
16. Navigate to the **DMQS** space, via the **Explore** option, if required.

17. Click the **ABC Sales Data_SA3** app and select the **Employees** sheet. What can you see? Consider why.
18. Navigate back to the **Hub** and log *Ingrid* out.

Solution



Activity 4b: Omit a group of fields (QSoW)

Summary:

- Modify app-level security to omit groups of fields.
- Analyze sheet filter objects to verify the results.

Qlik Sense app: ABC Sales Data_SA2 / ABC Sales Data_SA3

Challenge

In the **ABC Sales Data_SA3** app, omit a group of fields by doing the following:

- Create a copy of the *DataReductionSA.txt* and rename it to *DataReductionSA_OPT.txt*.
- In the new file, replace *OMIT* with *OMIT_GROUP* and all the *JobTitle* instances with *OMIT_GROUP1*.
- In the **Qlik Sense Hub**, open the **Everyone** stream and duplicate the **ABC Sales Data_SA2** app.
- Change the name to *ABC Sales Data_SA3*.
- In the **Security** script section, modify the **SECTION ACCESS LOAD** statement to load from the new file.

- Add a second **LOAD** statement containing table columns *OMIT_GROUP* and *OMIT*, and fields *JobTitle*, *SalesTitle*, and *EmploymentType*, associated in the same group, *OMIT_GROUP1*.
- Load your data. Publish the app using the **Qlik Management Console (QMC)** in the **Everyone** stream.
- Log on as *IHX*. Observe the fields the user has access to in the **Employees** sheet.

Hints

- The file that contains the security information is stored in the following location: **C:\Qlik Sense Data\TextFiles**.

Step-by-step guide

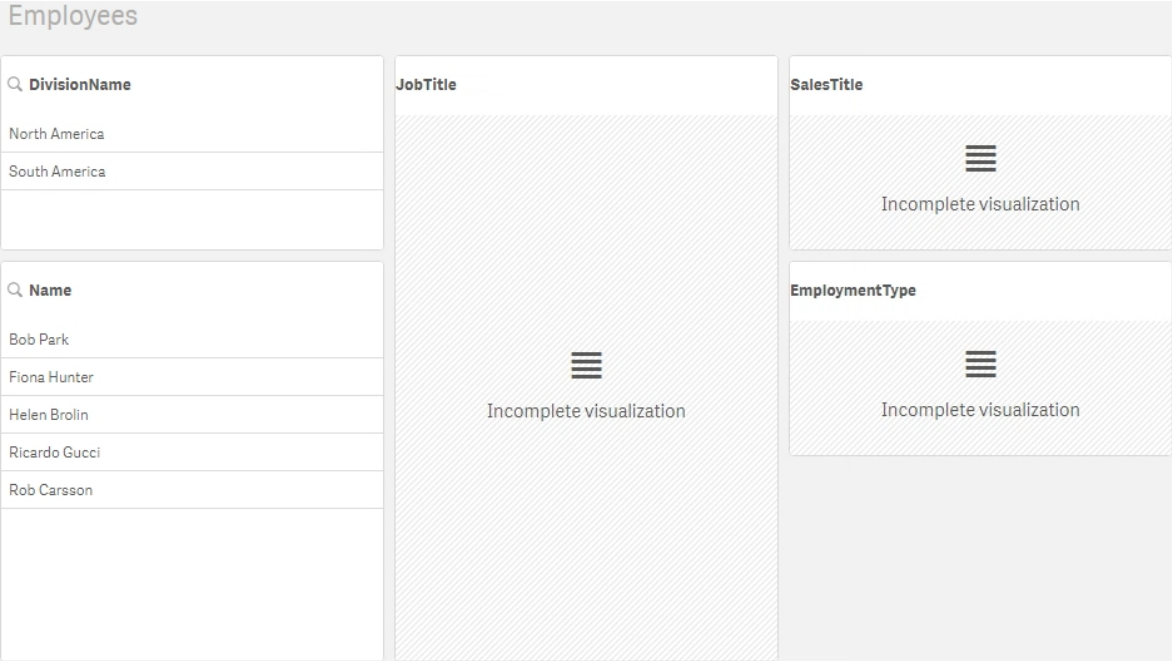
1. Examine the *DataReductionSA.txt* you loaded in the previous activity. This file can be found under:
   C:\Qlik Sense Data\TextFiles
2. Duplicate this file and rename the copy as *DataReductionSA_OPT.txt*. Open the file to edit it.
3. In the file, replace *OMIT* with *OMIT_GROUP*. Also, replace all the *JobTitle* instances with *OMIT_GROUP1*. Then, **Save** and close the file.
4. Open the **Qlik Sense Hub** and make sure you are logged on as *Student.* Go to the **Everyone** stream.
5. Duplicate the **ABC Sales Data_SA2** app. Rename the copy that appears in the **Work** area to *ABC Sales Data_SA3*.
6. Navigate to the **Data load editor** and locate the **Security** section.
7. Replace the name of the file *DataReductionSA.txt* with *DataReductionSA_OPT.txt* in the first **FROM** clause.
8. In the **SECTION ACCESS LOAD**, replace the *OMIT* field with *OMIT_GROUP*.
9. Between the **SECTION ACCESS** table and the **SECTION APPLICATION** statement, add the following script:

```
LOAD * INLINE [
   OMIT_GROUP, OMIT
   OMIT_GROUP1, JobTitle
   OMIT_GROUP1, SalesTitle
   OMIT_GROUP1, EmploymentType
];
```

10. Load the data to add the security information to the app and open the **QMC**.
11. In the **Apps** page, select and publish the **ABC Sales Data_SA3** app to the **Everyone** stream.
12. From the Windows **Start** button, right-click on the **Google Chrome** icon, and select **Run as different user**.
13. Log on as *Ingrid Hendrix*, using *IHX* as the userID and *Training1* as the password.
14. When **Google Chrome** opens, make sure you are viewing **https://qse/hub/**.
15. When the **Qlik Sense Hub** opens, confirm that the **IHX** credentials appear on the top bar. Select the **Everyone** stream.

16. Click the **ABC Sales Data_SA3** app and select the **Employees** sheet. What can you see?

Solution



Activity 5: Data profiling (SaaS)

Summary:

- Use built-in tools to profile and analyze data files.
- Verify data files and apps are in-sync.
- Add metadata to data files.

Qlik Sense app: ABC Sales Data

Challenge

Use the tools available in the **Hub** to profile data and apply metadata properties:

- Analyze the *Employees.qvd* file, using the **Profile data** option.
- Verify aggregate calculations.
- Add and remove metadata **properties** to the file.
- Review the data **lineage** of the *Employees* table.
- Analyze the *OrderDetails.qvd* file, using the **Profile data** option.
- Ensure a statistical match between the profile results and the **ABC Sales Data** app.

Hints

- *Employees.qvd* and *OrderDetails.qvd* are generated by running the data load script of the **ABC Sales Data** app.

Step-by-step guide

1. Ensure you are logged in to **Qlik Cloud** with your primary **Qlik ID**.
2. Navigate to the **Personal** space and select the **Data** from the **Types** filter.
3. Select the *Employees.qvd* file to open it. The page updates to display the **Overview** details of the data file.
4. Use the profile refresh icon, next to the button labeled **Create analytics app**, to ensure the profile data is up-to-date.
5. Observe and analyze the headline information for the table the file contains.
6. Click **View fields** to expand the display to show all fields in the table.
7. Locate the field of *Team* and observe that the four most frequent values are: *Sales, IT,Marketing*, and *Delivery*.
8. Click on the tile for the *DateOfBirth* field to display an additional **Details** panel.

   Note: If the fields are not displayed in **Tile** view, use the view controls on the right to change it.

9. Click the check box for **Personal Information** to mark the *DateOfBirth* field as containing such, then close the panel.
10. Navigate from the **Fields** tab to the **Overview** tab to return to the headline profile information for the *Employees.qvd* file.
11. Observe in the summary information at the top; the **Metadata modified** indicator has updated to reflect the recent change.
12. Click the **More actions** ellipsis button in the top right and select **Lineage**.
13. Move the view around by clicking and dragging, or zooming in and out as required to display the full lineage of this table. Observe that there were sources of data from **Qlik Data Files (Personal)** loaded into the **ABC Sales Data** app, which subsequently generated this downstream *Employees.qvd* file, residing within the same **File storage** of the **Personal** space.
14. Use the drop-down arrow on the **File storage** box on the left, to expand the sources.
15. Verify that there were at least four data file sources which were ingested by the **ABC Sales Data** app, in order to store the *Employees.qvd* file.
16. To return to the **Analytics Services** area, click on the Qlik logo in the top left.
17. As before, navigate to the **Personal** space and select the **Data** from the Type filter.
18. Select the *OrderDetails.qvd* file to open it.
19. Make sure you are viewing the **Overview** information.
20. Click **View fields**.
21. Ensure the view selector is set to **List view**.
22. Use the **Column picker** button in the top right of the table, and ensure the **Sum** aggregation is selected.

23. Check the **Sum** calculation for the *LineSalesAmount* field, and make a note of it.
24. Use the Qlik logo again to return to the **Analytics Services** and navigate back to the **Personal** space.
25. Locate the open the **ABC Sales Data** app.
26. Navigate to the **Bonus** sheet.
27. Ensure no selections are currently applied, and verify that the total *Sales* figure matches up with the figure being recently exported to the *OrderDetails.qvd*.

Activity 6: Reloading from the Hub (SaaS)

Summary:

- Initiate app reloads from the Hub.
- Analyze reload logs.
- Set up reload schedules.

Qlik Sense app: ABC Sales Data

Challenge

Investigate how reloads can be initiated from the **Hub**:

- Perform an interactive reload from the **Data load editor**.
- Observe the app reload **History**.
- Set up a reload schedule for the **ABC Sales Data** app.
- Alter a reload schedule and analyze a reload log.

Hints

- Do not unintentionally leave a recurring reload schedule running on an app, as it consumes resources.

Step-by-step guide

1. Ensure you are logged in to **Qlik Cloud** with your primary **Qlik ID**.
2. Navigate to the **Personal** space and use the **Types** filter to limit the display to just **Apps**.
3. Open the **ABC Sales Data** app and visit the **Data load editor**.
4. Use the **Load data** button to execute an interactive reload of the app data model.
5. When finished, close the **Data load progress** dialog box, then return to the **Hub**.
6. Use the **More actions** menu for the **ABC Sales Data** app to open the menu, and select **Details**.
7. Observe the **Last reload date** for the app as being the current date and time.
8. Switch to the **Reload history** tab.

9. Note that the history is empty.

> **Note:** Interactive reloads from the **Data load editor** do not get logged in the **Reload history**.

10. Use the **More actions** ellipsis to open the menu, and select **Reload now**.
    A notification appears at the top to indicate reload of the app has started. Once executed, another notification appears to state it is complete.
    An entry appears in the **Reload history** log with the relevant details.
11. Click the **View** button to review the new entry in the **Reload history** log.
12. Observe that while most of the log is an echo of what can be witnessed when performing an interactive reload, there is a single line at the top which indicates the **ReloadID** associated with this occurrence of a reload for the app.
13. **Close** the **Reload summary** dialog box.
14. Use the **More actions** ellipsis again, to open the menu. This time select **Reload schedule**.
15. Click **Add schedule**.
16. Change the **Reload this app** drop-down to *Yearly*.
17. Analyze the various options for setting repetition.
18. Select the prior month, and deselect this month, so that the next scheduled execution will take place 11 months from now.
19. **Save** the **Reload schedule** to set it, and return to the App **Reload history** page.

> **Note:** Tenant owners and admins can display all scheduled reloads for the tenant, as well as manage them to some degree.

20. While you have no access to the **Management Console** in the training environment, to edit the **Reload schedule** of an app, locate the **ABC Sales Data** app, and use the ellipsis actions button.
21. Select **Reload schedule**.
22. Because the app already has a reload schedule set, the current settings are displayed in the dialog box and can be altered. Use the **Delete** option to cancel the reload schedule.
23. Click **Save** to update the app reload schedule.
24. Use the **Back** button to return to the **Catalog**.

Activity 7: Evaluating app performance (SaaS)

Summary:

- Initiate app performance evaluations.
- Analyze the evaluation results.
- Understand the impact of Section Access security with respect to performance analysis.

Qlik Sense app: ABC Sales Data / ABC Sales Data_eval / ABC Sales Data_SA3

Challenge

Use the built-in SaaS app performance evaluation feature, to:

- Analyze app memory consumption.
- Understand how different tables and fields consume different levels of resource.
- Identify how to dig deeper into the analysis results.
- Appreciate the impact which **SECTION ACCESS** security has on performance evaluation.

Hints

- The performance evaluation tool is a feature to help guide Developers to potential issues with app development.

Step-by-step guide

1. Ensure you are logged in to **Qlik Cloud** with your primary **Qlik ID**.
2. Navigate to the **Personal** space and use the **Types** drop-down to filter by **Apps**, if required.
3. Duplicate the **ABC Sales Data** app.
4. Use the **More actions** ellipsis and the **Rename** option, to rename the copy to *ABC Sales Data_Eval*.
5. Click the **Save** button, then utilize **More actions** for the new app and select **Details**.
6. Select the **Performance evaluation** tab.
7. The list should be empty, so click **Evaluate performance** to queue up an app health check.
8. After a short duration, the evaluation should be complete, and an entry appears in the list, marked as **Complete**.
9. Click on the new entry to investigate the findings.
10. The evaluation dialog displays an **Overview** of the findings, indicating there is just one public sheet in the app, containing ten objects which it evaluated. Switch to the **Details** tab.
11. Expand **Top 5 tables with the largest memory allocation**.
12. The findings indicate that the *OrderDetails* table consumes the most memory of all tables, followed by the *Orders* table. This is logical, given the nature of these two tables as fact detail tables. Collapse the view.
13. Expand **Top 5 fields with the largest memory allocation**.
14. According to the analysis, the field *%OrderKey* accounts for the greatest memory allocation in the data model. This also makes sense, as the field is a deliberate creation to hold unique values, despite the use of the **AutoNumber** function. Collapse the view.
15. Expand **Top 5 objects with the largest cached load time**.
16. The table displays a list of object IDs and the associated time to load them. For the top item, click the button to **Open in new tab**.
    A new browser tab opens and displays the publicly-available **Employees** sheet. The relevant object is briefly indicated with a colored-border to draw attention to it.
17. Close the browser tab to return to the evaluation details. Then, click the **Done** button to close the dialog.
18. Go **Back** to the app list and select the **DMQS** space.

19. Locate the **ABC Sales Data_SA3** app with your own username suffix, click the **More actions** menu and go into the **Details**.
20. Select the **Performance evaluation** tab.
21. Click **Evaluate performance** to queue it up.
22. Once it completes, click on the new entry to open the details.
23. Observe the details on the **Overview** sheet.

    Note: The app still contains one public sheet with ten public objects to evaluate. However, the **SECTION ACCESS** utilized within the app can hamper the evaluation of the performance, as it is no longer a true reflection of the different data users may see.

24. Use the **Done** button to close the log, and the **Back** link to return to the **Hub**.

Activity 8 - Optional: Qlik Sense with R Demonstration

Summary:

- Explore the Qlik Sense Advanced Analytics Integration functionality.
- Leverage the Qlik demos site.

Qlik Sense app: Demo site | UK Crime

Challenge

In the **Qlik Demos** website, review how R is utilized and how it works in an app.

- Access the Qlik Sense demo site (https://demos.qlik.com/qliksense) and login with your Qlik Account.
- Select the *Advanced Analytics Integration* tag and select the **UK Crime** Demo app.
- Launch the Demo app and identify the sheets/charts where R is used.
- Access **Qlik Developer** (https://qlik.dev/changelog/60-api-data-connections-dcaas) and review the information on *Analytic Data Connections*.

Hints

- Charts that use R look very similar to Qlik Sense standard visualizations, but they are customized for a specific purpose.

Step-by-step guide

1. In a browser window, access the **Qlik Demos** site (https://demos.qlik.com/qliksense). Login with your Qlik account credentials. (If you do not have a login, select **Login** from the top navigation and, in the login window, select **Register**.)
2. Use the **Tags** drop-down to select **Advanced Analytics Integration**. This filters the demos that make use of the **Advanced Analytics Toolbox**.

3. Select **UK Crime**. Then, click **Launch Demo** to review the app.
4. Explore the different sheets in the app and identify the visualizations that contain **Advanced Analytics** expressions.
   Hint: Read the explanatory comments on every sheet.
5. Navigate to the **Crime Forecasts** sheet and adjust the slider to increase or decrease the *Forecast Months* value. Observe the line chart adjusting accordingly.
6. Navigate back to the **App overview**. Right-click and **Duplicate** the **Crime Forecasts** sheet. Open and edit the sheet you just created.

   Note: For the **Duplicate** option to appear when you right-click, make sure you are logged on with your Qlik account.

7. From the **Assets** panel, navigate to the **Master items** and then, **Measures**. Observe the list of all the Master Measures. The line chart uses three of these Measures: *HW Forecast*, *HW Lower 80%* and *HW Upper 80%*. Click to see their expressions. They are all using R script.

   Note: The purpose of this lesson is to demonstrate how we can use R expressions from within Qlik Sense. However, the explanation of the R syntax and what they do is beyond the scope of this course.

8. Navigate to the **Clustering by Crime Type** sheet and **Duplicate it**. In edit mode, review the dimensions and measures of the scatter plot. Notice that neither the dimensions nor the measures of the chart use R syntax. To achieve clustering the *Crime Types* using the K-means clustering algorithm, the R expression is used to color the data points.
9. In the **Properties** panel, navigate to the **Colors and legend** section under **Appearance**. Notice that the *Crime Percentage and Resolution Time Clusters* Master Measure is used to color the data points. From the **Assets** panel, find the Master Measure and review its expression. R syntax is used to distinguish the crime types that belong to the same cluster by using the same color.
10. The expression used in this Master Measure is dynamic. We can use the **Clusters** drop-down menu at the top right of the sheet to change the number of clusters required for our analysis. Navigate to the **App overview** window to stop editing the sheet.
11. Similarly, navigate to the other app sheets and identify other objects with **Advanced Analytics** expressions.
12. In a new browser window, access the Qlik Dev site (https://qlik.dev/changelog/60-api-data-connections-dcaas).
13. Use the links within the page to navigate to and learn more about **analytic data connections**.

    Note: These solutions are provided by Qlik developers free of charge. Download them from the Qlik Sense Demos site and start using them in your app.